

Momit

- 4) Consider the following relational schema and briefly answer the questions that follow:
- Emp(*eid*: integer, *ename*: string, *age*: integer, *salary*: real)
 Works(*eid*: integer, *did*: integer, *pct time*: integer)
 Dept(*did*: integer, *budget*: real, *managerid*: integer)
- Write the SQL statements required to create these relations, including all integrity constraints such as primary and foreign keys.
 - Define a domain constraint on Emp that will ensure that every employee makes at least \$10,000.
 - Define an assertion on Dept that ensure that all managers have *age* > 30.
1. (b) Define referential integrity. Explain the tests that must be made to preserve referential integrity for *delete* operation.
2. Suppose you are given a relation *R* with four attributes *ABCD*. Foreach of the following sets of FDs, assuming those are the only dependencies that holdfor *R*, do the following:
- Identify the candidate key(s) for *R*.
 - Identify the types of functional dependencies (FD) that exist. Why do you think those are FDs?
 - Identify the bestnormal form that *R* satisfies (1NF, 2NF, 3NF, or BCNF).
 - If *R* is not in BCNF, decompose it into a set of BCNF relations that preserve the dependencies.
 - $C \rightarrow D, C \rightarrow A, B \rightarrow C$
 - $B \rightarrow C, D \rightarrow A$
 - $ABC \twoheadrightarrow D, D \rightarrow A$
 - $A \twoheadrightarrow B, BC \twoheadrightarrow D, A \twoheadrightarrow C$
 - $AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B$

OR

2. When a relational schema will be in 1NF, 2NF, and 3NF. Illustrate with an example how you can convert the following Schema *Sales Record* into 3NF.10

CustName	Item	ShippingAddress	Newsletter	Supplier	SupplierPhone	Price
Hasan	Galaxy	3/1 Ctg	Gal-News	Samsung	22-8805568	25000
Karwal	iPhone	2-A Dhk	I-News	Apple	66-5668412	65000
Sojib	Galaxy, iPhone	255 Raj	Gal-News, I-News	Wholesale	Toll-Free	90000
Hasan	iPhone	1/6A Syl	I-News	Apple	66-5668412	65000

Ans to ques NO 1 (a)

1

```
CREATE TABLE Emp ( eid INTEGER,  
                    ename CHAR(10),  
                    age INTEGER,  
                    salary REAL,  
                    PRIMARY KEY (eid) )
```

```
CREATE TABLE Works (eid INTEGER,  
                    did INTEGER,  
                    pcttime INTEGER,  
                    PRIMARY KEY(eid, did),  
                    FOREIGN KEY(did) REFERENCES Dept,  
                    FOREIGN KEY(eid) REFERENCES Emp, ON DELETE CASCADE)
```

```
CREATE TABLE Dept ( did INTEGER,  
                    buget REAL,  
                    managerid INTEGER,  
                    PRIMARY KEY(did),  
                    FOREIGN KEY(managerid) REFERENCES Emp, ON DELETE SET NULL)
```

2

```

CREATE TABLE Emp (eid INTEGER,
                   ename CHAR(10),
                   age INTEGER,
                   salary REAL,
                   PRIMARY KEY (eid),
                   CHECK (salary >= 10000)
)

```

3

```

CREATE ASSERTION managerAge
CHECK ((SELECT E.age
        FROM Emp E, Dept D
        WHERE E.eid = D.managerid ) > 30 )

```

Ans to the ques No 1(b)

Ans to ques 1(b)

Referential integrity:- It is concerned with the concept of foreign key.

It creates relation between more than one table.

The rule for the referential integrity is that in one table referential attribute must be primary key and in the other it must be present not necessarily a primary key

ID	Name	Age	DID
1	Mohit	50	87
2	Salim	60	101
3	Shihab	70	102

reference.

DID	Name	Course
103	Khalid	CSE-12
87	Mohit	CSE-13
102	Shihab	CSE-15

Primary Key

For Delete Operation

During a delete operation, if a row to be deleted contains a value that is referenced by a foreign key in another table, then the referential integrity can be avoided by either of the two ways:

1. Deleting the original row and also deleting the referencing row from the other table. This action is called **cascaded Deletion** and can cascade through a series of referencing tables.
2. Deleting neither of the rows i.e. taking no action. This action is called **Restricted Deletion**.

Ans to the ques No 2(b)

There are mainly four types of Functional Dependency in DBMS. Following are the types of Functional Dependencies in DBMS:

- **Multivalued Dependency**
- **Trivial Functional Dependency**
- **Non-Trivial Functional Dependency**
- **Transitive Dependency**

Multivalued Dependency in DBMS

Multivalued dependency occurs in the situation where there are multiple independent multivalued attributes in a single table. A multivalued dependency is a complete constraint between two sets of attributes in a relation. It requires that certain tuples be present in a relation. Consider the following Multivalued Dependency Example to understand.

Trivial Functional Dependency in DBMS

The Trivial dependency is a set of attributes which are called a trivial if the set of attributes are included in that attribute.

So, $X \rightarrow Y$ is a trivial functional dependency if Y is a subset of X . Let's understand with a Trivial Functional Dependency Example.

Non Trivial Functional Dependency in DBMS

Functional dependency which also known as a nontrivial dependency occurs when $A \rightarrow B$ holds true where B is not a subset of A . In a relationship, if attribute B is not a subset of attribute A , then it is considered as a non-trivial dependency.

Transitive Dependency in DBMS

A Transitive Dependency is a type of functional dependency which happens when "t" is indirectly formed by two functional dependencies. Let's understand with the following Transitive Dependency Example.

Ans to Question 2

①

Ⓐ Candidate key: B

C/ R is in 2NF.

d/ $C \rightarrow D$ and $C \rightarrow A$ both cause violation of BCNF. One way to (lossless) join preserving decomposition is to decompose R into AC, BC, and CD.

②

① Candidate keys : BD

~~② R is 1~~

③ R is in 1NF. ~~but NOT~~

④ Both $B \rightarrow C$ and $D \rightarrow A$ cause BCNF violations. The decomposition AD, BC, BD (obtained by first decomposing to AD, BCD) is BCNF and lossless and join preserving

③
① Candidate keys: $ABCD, BCD$

② R is in 3NF

③ $ABCD$ is not BCNF since $D \rightarrow A$ and D is not a key. However if we split up R as AD, BCD we cannot preserve the dependency $ABC \rightarrow D$. So, there is no BCNF decomposition

④
① Candidate keys: A

② R is in 2NF. ~~but Not~~

③ $BC \rightarrow D$ violates BCNF since BC does not contain a key. So we split up R as in: BCD, ABC .

5

① Candidate keys AB, BC, CD, AD

② R is in 3NF

③ $C \rightarrow A$ and $D \rightarrow B$ both cause violations, so decomposition into AC, BCD. but this does not preserve $AB \rightarrow C$ and $AB \rightarrow D$, and BCD is still not BCNF because $D \rightarrow B$. So we need to decompose further into AC, BD, CD. However when we attempt to revive the lost functional dependencies by adding ABC and ABD, we find that these relations are not in BCNF form. So, there is no BCNF decomposition.

2 con

Converting the given schema sales
Record into 3NF

~~We~~
At first we have to transform
the schema into 1NF

Cust Number	Item	Shipping Add	News id	Supplier	Supplier Phone	Price
Hasan	Galaxy	3/1 Ctg	Crat-News	Samsung	2290-	25000
Kamal	iphone	2A Dhk	i-news	Apple	6656	65000
Sajib	Cat Galaxy	255 Raj	Crat-news	Wholesale	Toll-free	25000
Sajib	iphone	255 Raj	i-news	Wholesale	Toll-free	65000
Hasan	iphone	1/6 ASyd	Crat-news	Apple	6656	65000

Now, 2DFNF:

Table 1: "User info" Customer-record

Cust Name	Shipping Add	Supplier	Supplies phone
Hasan	Galaxy	3/1 Ctg	
Hasan	3/1 Ctg	Samsung	22-88 - - - -
Kamal	2-A Dhk	Apple	66-566 - - - -
Sojib	255 Raj	Wholesale	Toll-free
Hasan	1/6A Syl	Apple	66-566

Table 2: "Phone-info"

Cust Name	Item	Newsletter	Price
Hasan	iphone		
Hasan	Galaxy	Gal-News	25,000
Kamal	iphone	i-news	65,000
Sojib	Galaxy	Galnews	toll-free 25,000
Sojib	iphone	i-news	65,000
Hasan	iphone	i-news	25,000

Now, to finally we have to just
 remove a transitive dependency from
 Table 2.

So,

Now,

Cust Name	Item
Hasan	Galaxy
Kamal	iphone
Sojib	Galaxy
Sojib	iphone
Hasan	iphone

Another table will be let's name
 it 'item-info'

Item	Newsletter	Price
Galaxy	Gal-News	25,000
iphone	ipi-news	65,000